

Keep it on a separate partition

If your backup directory is on a separate partition, then any corruption in the main filesystem will not normally affect the backup. If the backup process runs out of disk space, it will fail, but it won't take the rest of the system down too. More importantly, keeping your backups on a separate partition means you can keep them mounted read-only; we'll discuss that in more detail in the next chapter.

Keep that partition on a separate disk

If your backup partition is on a separate hard disk, then you're also protected from hardware failure. That's very important, since hard disks always fail eventually, and often take your data with them. An entire industry has formed to service the needs of those whose broken hard disks contained important data that was not properly backed up.

Important: Notice, however, that in the event of hardware failure you'll still lose any changes made since the last backup. For home or small office users, where backups are made daily or even hourly as described in this document, that's probably fine, but in situations where any data loss at all would be a serious problem (such as where financial transactions are concerned), a RAID system might be more appropriate.

RAID is well-supported under Linux, and the methods described in this document can also be used to create rotating snapshots of a RAID system.

Keep that disk on a separate machine

If you have a spare machine, even a very low-end one, you can turn it into a dedicated backup server. Make it standalone, and keep it in a physically separate place – another room or even another building. Disable every single remote service on the backup server, and connect it only to a dedicated network interface on the source machine.

On the source machine, export the directories that you want to back up via read-only NFS to the dedicated interface. The backup server can mount the exported network directories and run the snapshot routines discussed in this article as if they were local. If you opt for this approach, you'll only be remotely vulnerable if:

1. a remote root hole is discovered in read-only NFS, and
2. the source machine has already been compromised.

We'd consider this "pretty good" protection, but if you're (wisely) paranoid, or your job is on the line, build two backup servers. Then you can make sure that at least one of them is always offline.

If you're using a remote backup server and can't get a dedicated line to it (especially if the information has to cross somewhere insecure, like the public internet), you should probably skip the NFS approach and use `rsync`